

● HARD

● ADVANCED MATH

● ANSWER KEY

Nonlinear functions

08

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Q1**D**

D. $60,000 = 50,000(1.03)^t$

Choice D is correct. Stating that the population will increase each year by 3% from the previous year is equivalent to saying that the population each year will be 103% of the population the year before. Since the initial population is 50,000, the population after t years is given by $50,000(1.03)^t$. It follows that the equation $60,000 = 50,000(1.03)^t$ can be used to estimate the number of years it will take for the population to reach 60,000.

Choice A is incorrect. This equation models how long it will take the population to decrease from 60,000 to 50,000, which is impossible given the growth factor. Choice B is incorrect and may result from misinterpreting a 3% growth as growth by a factor of 3. Additionally, this equation attempts to model how long it will take the population to decrease from 60,000 to 50,000. Choice C is incorrect and may result from misunderstanding how to model percent growth by multiplying the initial amount by a factor greater than 1.

Q2

C

C. 36

Choice C is correct. For a function of the form $fx = ar^{\frac{x}{k}}$, where a , r , and k are constants and $r < 1$, the value of fx decreases by $100(1 - r)\%$ for every increase of x by k . In the given function, $a = 5,470$, $r = 0.64$, and $k = 12$. Therefore, for the given function, the value of fx decreases by $100(1 - 0.64)\%$, or 36% , for every increase of x by 12. Since fx represents the value, in dollars, of the equipment after x months of use, it follows that the value of the equipment decreases every 12 months by 36% of its value the preceding 12 months. Since there are 12 months in a year, the value of the equipment decreases each year by 36% of its value the preceding year. Thus, the value of p is 36.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q3

20

The correct answer is 20. It's given that $fx = -a^x + b$. Substituting $-a^x + b$ for fx in the equation $y = fx - 12$ yields $y = -a^x + b - 12$. It's given that the y -intercept of the graph of $y = fx - 12$ is 0, $-\frac{75}{7}$. Substituting 0 for x and $-\frac{75}{7}$ for y in the equation $y = -a^x + b - 12$ yields $-\frac{75}{7} = -a^0 + b - 12$, which is equivalent to $-\frac{75}{7} = -1 + b - 12$, or $-\frac{75}{7} = b - 13$. Adding 13 to both sides of this equation yields $\frac{16}{7} = b$. It's given that the product of a and b is $\frac{320}{7}$, or $ab = \frac{320}{7}$. Substituting $\frac{16}{7}$ for b in this equation yields $a\frac{16}{7} = \frac{320}{7}$. Dividing both sides of this equation by $\frac{16}{7}$ yields $a = 20$.

Q4

B

B. $-\frac{7}{4}, 0$

Choice B is correct. Since the line of symmetry for the graph of a quadratic function contains the vertex of the graph, the x -coordinate of the vertex of the graph of $y = f(x)$ is the x -coordinate of the midpoint of its two x -intercepts. The midpoint of two points with x -coordinates x_1 and x_2 has x -coordinate x_m , where $x_m = \frac{x_1 + x_2}{2}$. It's given that the vertex is $-3, 6$ and one of the x -intercepts is $-\frac{17}{4}, 0$. Substituting -3 for x_m and $-\frac{17}{4}$ for x_1 in the equation $x_m = \frac{x_1 + x_2}{2}$ yields $-3 = \frac{-\frac{17}{4} + x_2}{2}$. Multiplying each side of this equation by 2 yields $-6 = -\frac{17}{4} + x_2$. Adding $\frac{17}{4}$ to each side of this equation yields $-\frac{7}{4} = x_2$. Therefore, the other x -intercept is $-\frac{7}{4}, 0$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q5

D

D. -25

Choice D is correct. Substituting b for x in the given equation yields

$g(b + 28) = \frac{1}{10}(b - a)^2(b - b)^2(b - c)$, which is equivalent to

$g(b + 28) = \frac{1}{10}(b - a)^2(0)^2(b - c)$, or $g(b + 28) = 0$. It follows that when $x = b + 28$, $g(x) = 0$

. There are three points where the graph of $y = g(x)$ intersects the x -axis: $(-2, 0)$, $(1, 0)$, and

$(3, 0)$. Therefore, $g(-2) = 0$, $g(1) = 0$, and $g(3) = 0$. It follows that either $b + 28 = -2$,

$b + 28 = 1$, or $b + 28 = 3$. Subtracting 28 from each side of the equation $b + 28 = -2$ yields

$b = -30$. Subtracting 28 from each side of the equation $b + 28 = 1$ yields $b = -27$. Subtracting

28 from each side of the equation $b + 28 = 3$ yields $b = -25$. Thus, the value of b must be

either -30 , -27 , or -25 . Of these values, only -25 is given as a choice.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This would be a possible value of b if the graph shown was the graph of $y = g(x + 28)$, not $y = g(x)$.

Choice C is incorrect. This would be the value of c , not the value of b , if the graph shown was the graph of $y = g(x + 28)$, not $y = g(x)$.

Q6**B****B.** -9

Choice B is correct. It's given that $gx = fx - 1$. Since $fx = x + 6x + 5x + 1$, it follows that $fx - 1 = x - 1 + 6x - 1 + 5x - 1 + 1$. Combining like terms yields $fx - 1 = x + 5x + 4x$. Therefore, $gx = xx + 5x + 4$. The x -intercepts of a graph in the xy -plane are the points where $y = 0$. The x -coordinates of the x -intercepts of the graph of $y = gx$ in the xy -plane can be found by solving the equation $0 = xx + 5x + 4$. Applying the zero product property to this equation yields three equations: $x = 0$, $x + 5 = 0$, and $x + 4 = 0$. Solving each of these equations for x yields $x = 0$, $x = -5$, and $x = -4$, respectively. Therefore, the x -intercepts of the graph of $y = gx$ are $0, 0, -5, 0$, and $-4, 0$. It follows that the values of a , b , and c are $0, -5$, and -4 . Thus, the value of $a + b + c$ is $0 + -5 + -4$, which is equal to -9 .

Choice A is incorrect. This is the value of $a + b + c$ if $gx = fx + 1$.

Choice C is incorrect. This is the value of $a + b + c - 1$ if $gx = x - 6x - 5x - 1$.

Choice D is incorrect. This is the value of $a + b + c$ if $fx = x - 6x - 5x - 1$.

Q7**-4.9**Accept also: $-49/10$

The correct answer is -4.9 . The value of $f-4$ is the value of fx when $x = -4$. Substituting -4 for x in the equation $fx = 3 + -2x - x^2$ yields $f-4 = 3 + -2(-4) - (-4)^2$, or $f-4 = 3 + -8$, which is equivalent to $f-4 = 3 + 8$, or $f-4 = 11$. Since it's given that $f-4 = c$, it follows that $c = 11$ and the value of gc is the value of $g11$. Substituting 11 for w in the equation $gw = \frac{-w}{w-1} - w + 5$ yields $g11 = \frac{-11}{11-1} - 11 + 5$, or $g11 = -1.1 - 6$, which is equivalent to $g11 = 1.1 - 6$, or $g11 = -4.9$. Note that -4.9 and $-49/10$ are examples of ways to enter a correct answer.

Q8**B**

B. $n = 722.5^t$

Choice B is correct. Since the model estimates that the number of squirrels in the population increased by a fixed percentage, 150%, each year, the model can be represented by an exponential equation of the form $n = a1 + \frac{p}{100}^t$, where a is the estimated number of squirrels in the population at the end of 2015, and the model estimates that at the end of each year, the number is $p\%$ more than the number at the end of the previous year. Since the model estimates that at the end of each year, the number was 150% more than the number at the end of the previous year, $p = 150$. Substituting 150 for p in the equation $n = a1 + \frac{p}{100}^t$ yields $n = a1 + \frac{150}{100}^t$, which is equivalent to $n = a1 + 1.5^t$, or $n = a2.5^t$. It's given that the estimated number of squirrels at the end of 2016 was 180. This means that when $t = 1$, $n = 180$. Substituting 1 for t and 180 for n in the equation $n = a2.5^t$ yields $180 = a2.5^1$, or $180 = 2.5a$. Dividing each side of this equation by 2.5 yields $72 = a$. Substituting 72 for a in the equation $n = a2.5^t$ yields $n = 722.5^t$.

Choice A is incorrect. This equation represents a model where at the end of each year, the estimated number of squirrels was 150% of, not 150% more than, the estimated number at the end of the previous year.

Choice C is incorrect. This equation represents a model where at the end of each year, the estimated number of squirrels was 150% of, not 150% more than, the estimated number at the end of the previous year, and the estimated number of squirrels at the end of 2015, not the end of 2016, was 180.

Choice D is incorrect. This equation represents a model where the estimated number of squirrels at the end of 2015, not the end of 2016, was 180.

Q9**76**

The correct answer is 76. It's given that the graph of $y = gx$ is the result of translating the graph of $y = fx$ up 4 units in the xy -plane. It follows that the graph of $y = gx$ is the same as the graph of $y = fx + 4$. Substituting gx for y in the equation $y = fx + 4$ yields $gx = fx + 4$. It's given that $fx = x - 6x - 2x + 6$. Substituting $x - 6x - 2x + 6$ for fx in the equation $gx = fx + 4$ yields $gx = x - 6x - 2x + 6 + 4$. Substituting 0 for x in this equation yields $g0 = 0 - 60 - 20 + 6 + 4$, or $g0 = 76$. Thus, the value of $g0$ is 76.

Q10

B

B. $3 + 2\frac{55^2}{4}$

Choice B is correct. It's given that function g is defined by $g(t) = f(t) + 3$ and that $f(t) = 55t - 2t^2$. Substituting $55t - 2t^2$ for $f(t)$ in $g(t) = f(t) + 3$ yields $g(t) = 55t - 2t^2 + 3$, or $g(t) = -2t^2 + 55t + 3$. The maximum value of $g(t)$ can be found by completing the square to rewrite the equation defining g in the form $g(t) = a(t - h)^2 + k$, where the maximum value of the function is k , which occurs when $t = h$, and a is a negative constant. The equation $g(t) = -2t^2 + 55t + 3$ is equivalent to $g(t) = -2\left(t^2 - \frac{55}{2}t\right) + 3$, which can be rewritten as $g(t) = -2\left(t^2 - \frac{55}{2}t + \left(\frac{55}{4}\right)^2\right) + 3 + 2\left(\frac{55}{4}\right)^2$, or $g(t) = -2\left(t - \frac{55}{4}\right)^2 + 3 + 2\left(\frac{55}{4}\right)^2$. This equation is in the form $g(t) = a(t - h)^2 + k$, where $a = -2$, $h = \frac{55}{4}$, and $k = 3 + 2\left(\frac{55}{4}\right)^2$. Thus, the maximum value of $g(t)$ is $3 + 2\left(\frac{55}{4}\right)^2$.

Alternate approach: Since the function f is a quadratic function, the maximum value of $f(t)$ occurs at the value of t that's halfway between the two zeros of the function. The zeros of function f can be found by substituting 0 for $f(t)$ in the equation defining f , which yields $0 = 55t - 2t^2$. This equation can be rewritten as $0 = t(55 - 2t)$. By the zero product property, it follows that $t = 0$ or $55 - 2t = 0$. Subtracting 55 from each side of the equation $55 - 2t = 0$ yields $-2t = -55$. Dividing each side of this equation by -2 yields $t = \frac{55}{2}$. Therefore, the zeros of function f are 0 and $\frac{55}{2}$. The value that's halfway between 0 and $\frac{55}{2}$ can be found by calculating the average of 0 and $\frac{55}{2}$, which is $\frac{0 + \frac{55}{2}}{2}$, or $\frac{55}{4}$. It follows that the maximum of function f occurs when $t = \frac{55}{4}$. Substituting $\frac{55}{4}$ for t in the equation defining function f yields $f\left(\frac{55}{4}\right) = 55\left(\frac{55}{4}\right) - 2\left(\frac{55}{4}\right)^2$, which is equivalent to $f\left(\frac{55}{4}\right) = \frac{55^2}{4} - 2\left(\frac{55^2}{4^2}\right)$. Multiplying $\frac{55^2}{4}$ by $\frac{4}{4}$ in this equation to get a common denominator yields $f\left(\frac{55}{4}\right) = 4\left(\frac{55^2}{4^2}\right) - 2\left(\frac{55^2}{4^2}\right)$, or $f\left(\frac{55}{4}\right) = 2\left(\frac{55^2}{4^2}\right)$, which is equivalent to $f\left(\frac{55}{4}\right) = 2\left(\frac{55}{4}\right)^2$. Thus, the maximum value of $f(t)$ is $2\left(\frac{55}{4}\right)^2$. Since the equation defining $g(t)$ is $g(t) = f(t) + 3$, the maximum value of $g(t)$ is 3 greater than the maximum value of $f(t)$. It follows that the maximum value of $g(t)$ is $3 + 2\left(\frac{55}{4}\right)^2$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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